

Axis Hydraulics <
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W - Witness; H - Hold Point; S - Surveillance; X - Self Inspection

Drawing Number	HC-201 REV C	Area	Roof Plant Room — HWU-01 and HWU-02 — Roof level, Grid 6-8
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MATERIALS:			
Item:	Requirements:	Batch / cert. ref.	Check
Hot water pipework	Copper Type B to AS 1432 or PEX / PP-R rated for the design temperature and pressure	Press copper Type B 50/40, WM-020017	X
Fittings	Rated for continuous hot water service, WaterMark certified	Tempering valves 50°C, WM-022640	X
Tempering / TMV	Valves to AS 4032.2 / AS 4032.3, WaterMark certified	Expansion control valve, WM-022702	X
Insulation	Thickness to NCC Section J with continuous vapour and weather protection	Pipe insulation 25mm, Batch 24-0864	X
Expansion control	Expansion control valve and relief line to a safe, visible discharge	Valves and unions, WM-022118	X

INSPECTION & TEST PLAN				
No.	Installation:	Acceptance Criteria	Point	Initial & Date
1.0	Check current revision of drawings issued for construction, including any RFI or site instruction affecting this area.	In accordance with the project Drawing Register. Superseded revisions removed from site.	X	DS 08/09/2026
2.0	Set out heated water service against the coordinated services model / shop drawings. Confirm no clash with structure, ductwork, electrical or ceiling zones and that the design falls or grades can be achieved.	In accordance with the coordinated shop drawings and the hydraulic specification. Clashes resolved and reissued before installation.	X	DS 08/09/2026
3.0	Check materials delivered to site for compliance and damage (Copper Type B / PEX / PP-R rated for the design temperature). Record batch / lot numbers and WaterMark or certification details.	In accordance with AS/NZS 3500.2:2021 Section 2.3 'Selection and Use of Materials and Products', WaterMark certification and the hydraulic specification.	X	DS 08/09/2026
4.0	Check pipe support, bracketing and fixing centres, including provision for thermal movement and seismic restraint where specified.	In accordance with AS/NZS 3500.4:2021 Section 6 'Installation', manufacturer's requirements and the hydraulic specification. Fixings suitable for the substrate.	X	DS 08/09/2026
5.0	Check flow and return pipe sizes, circulation arrangement, balancing valves and dead leg lengths. Longest dead leg: 1.2 m	In accordance with AS/NZS 3500.4:2021 and the approved drawings. Dead legs within the maximum permitted length so delivery time is met.	X	DS 08/09/2026
6.0	Check provision for thermal expansion — expansion loops, offsets, anchors and guides.	In accordance with the pipe manufacturer's expansion data and the hydraulic specification.	X	DS 08/09/2026
7.0	Check tempering valve / TMV installation, orientation, isolation, strainers and access for servicing.	In accordance with AS/NZS 3500.4:2021 Section 1.9 and the manufacturer's instructions. Serviceable access provided.	X	DS 08/09/2026
8.0	Check expansion control valve, relief line and discharge point termination.	In accordance with AS/NZS 3500.4:2021. Relief discharge visible, unrestricted and to a safe location.	X	DS 08/09/2026
9.0	Check penetrations through fire and acoustic rated elements are sealed with a tested and approved system.	Fire collars / wraps installed in accordance with AS 4072.1 and the tested system, matching the FRL of the element penetrated. Acoustic seals in accordance with the acoustic report.	S	DS 08/09/2026
10.0	Check pipe insulation type, thickness and vapour sealing, including at supports and valves.	In accordance with NCC Section J and the hydraulic specification. Continuous, unbroken vapour barrier with no compression at supports.	X	DS 08/09/2026
11.0	Hydrostatic pressure test of the completed section to 1500 kPa (or 1.5 × maximum working pressure, whichever is the greater) held for 30 minutes. Test pressure held: 1500 kPa Released by M. Hartley (INSP-3368) 08/09/2026	No loss of pressure and no visible leakage over 30 minutes, in accordance with AS/NZS 3500.4:2021 and AS/NZS 3500.1:2021 Section 16 'Testing and Commissioning' and the hydraulic specification.	W	DS 08/09/2026
12.0	Flush pipework on completion of that section and confirm free discharge.	In accordance with the hydraulic specification. No debris, free flow observed at the downstream point.	X	DS 08/09/2026
13.0	Commission the heated water service and record delivery temperatures at the most remote and most disadvantaged outlets. Delivery temperature: 44.6 °C Note: Verified at the furthest ablation outlet after 60 seconds. Commissioning sheets attached to this ITP. Released by M. Hartley (INSP-3368) 08/09/2026	Delivery temperature not exceeding 45 °C at ablation fixtures in the applicable classes (50 °C elsewhere) in accordance with AS/NZS 3500.4:2021, and stored / circulated temperature to AS/NZS 3666.1 for Legionella control.	H	DS 08/09/2026
14.0	Apply pipe identification and directional flow markings to the heated water service.	Red-brown / "HOT WATER" with flow and return marked identification in accordance with AS 1345 and AS 2700, at intervals and locations required by the hydraulic specification.	X	DS 08/09/2026
15.0	Operate and label all isolation valves; confirm access, orientation and zone of control.	Valves accessible and operable, labelled to the valve schedule, and shown on the valve chart.	X	DS 08/09/2026

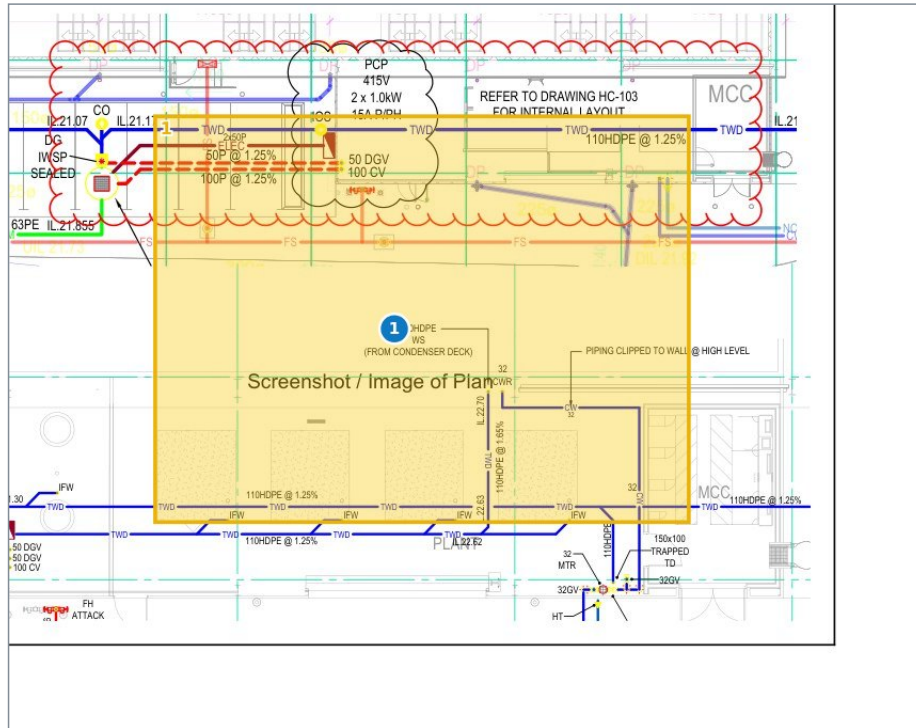
INSPECTION & TEST PLAN				
No.	Installation:	Acceptance Criteria	Point	Initial & Date
16.0	Final visual inspection of the completed installation; make good all damage, protective coatings and surrounds.	No visible defects. Area left clean and all protection removed. Defects register updated.	X	DS 08/09/2026
17.0	Record as-installed dimensions, invert levels and deviations for the as-built / work-as-executed drawings.	Marked-up drawings and survey pick-up submitted to the design team prior to covering.	X	DS 08/09/2026

INSTALLER SIGN-OFF Name: Darren Shoobridge Licence / CP no.: Date completed: 08/09/2026 Signature:  Date: 08/09/2026	CLIENT / SUPERINTENDENT ACCEPTANCE Name: M. Hartley Company: BESIX WATPAC 17 of 17 applicable items signed Signature:  Date: 08/09/2026
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PHOTOGRAPHIC RECORD

ITP 024 — Hot Water Service · Roof Plant Room — HWU-01 and HWU-02 · Roof level, Grid 6-8

Plan extract — HC-201 REV C · Level 3 hydraulic services — layout · 1 highlighted extent, 1 location marked



1. Yellow area — Roof plant room — hot water plant
1. HWU-01 flow and return connections (1 photo)

Photographs (3)



Item 1.0 · Installation
08/09/2026 04:45:44 (recorded)



Item 3.0 · Installation
08/09/2026 04:45:54 (recorded)



Pin 1 · Installation
08/09/2026 04:46:24 (recorded)